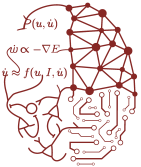
Asynchronous Integrated Circuits
An Introduction



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Wednesday 7th January, 2026

b
UNIVERSITÄT
BERN

Part I - INTEGRATED CIRCUITS

GUT meeting 13/01/2026

INTEGRATED CIRCUITS

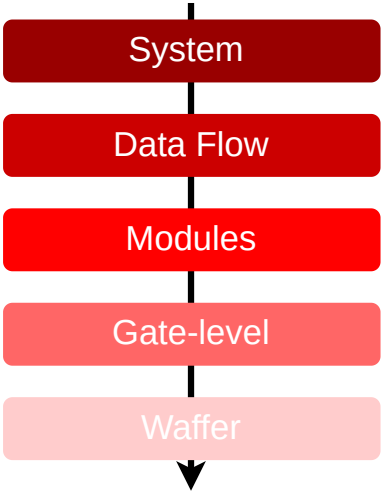
1. Abstractions
   - Layers
   - Combinational Logic
2. Design Families
3. Design Flow

ASYNCHRONOUS CIRCUITS

4. Linear pipeline or digital Fordism
5. Clock-driven
6. Muller Pipeline
7. Bundled-Data

Layers

Abstractions



System on Chip (SoC), Intellectual Properties (IPs), Memories, Inputs/Outputs (IOs)

Pipelines, Algorithms, Register Transfer Level (RTL)

Functional Blocks, Procedures

Logic Gates, Flip-Flops, Latches, Standard Cells

Masks

$$S = \bar{A}\bar{B}C_{in} + \bar{A}B\bar{C_{in}} + A\bar{B}\bar{C_{in}} + ABC_{in} \quad (1)$$

$$S = A \oplus B \oplus C_{in} \quad (2)$$

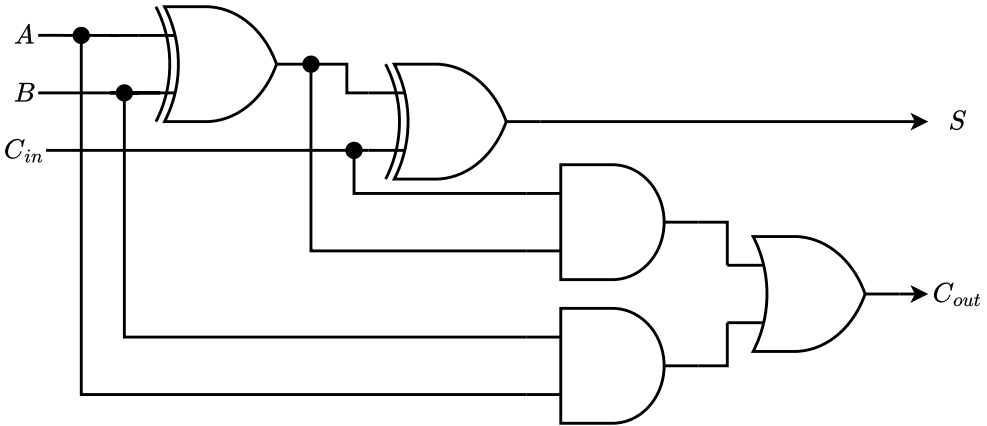
$$C_{out} = \bar{A}BC_{in} + A\bar{B}C_{in} + AB\bar{C_{in}} + ABC_{in} \quad (3)$$

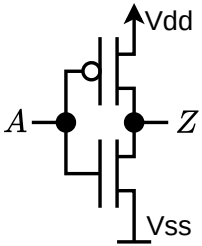
$$C_{out} = AB + C_{in}(A \oplus B) \quad (4)$$

Note

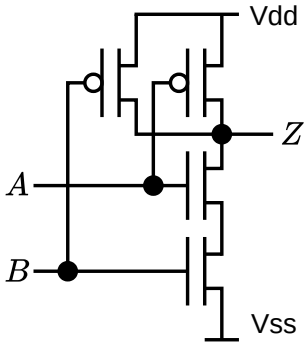
- (1) and (3) are in Disjunctive Normal Form (DNF)

Input			Output	
A	B	C_{in}	S	C_{out}
0	0	0	0	0
0	0	1	1	0
0	1	0	1	0
0	1	1	0	1
1	0	0	1	0
1	0	1	0	1
1	1	0	0	1
1	1	1	1	1

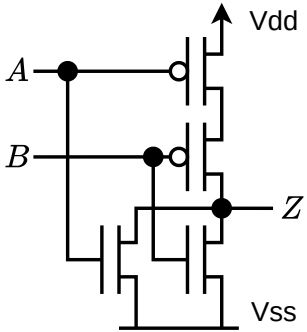

A
B
C_{in}
S
C_{out}



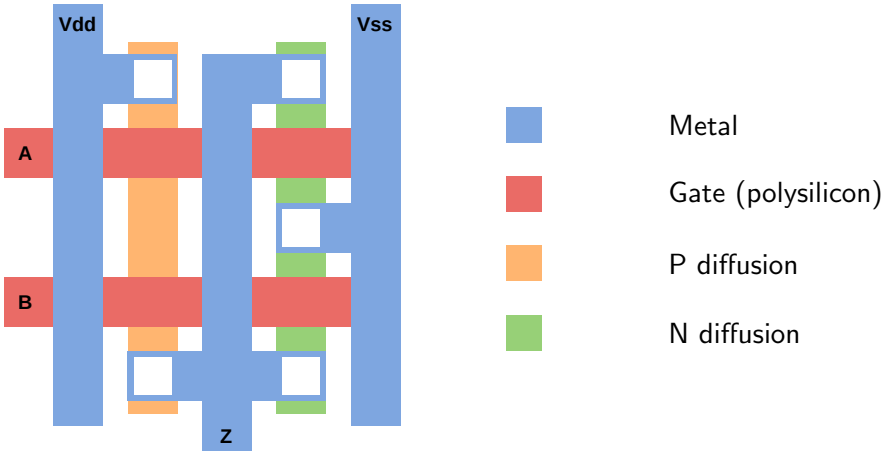
Inverter gate CMOS



NAND gate CMOS



NOR gate CMOS


Vdd
Vss
A
B
Z
Metal
Gate (polysilicon)
P diffusion
N diffusion

INTEGRATED CIRCUITS

1 Abstractions

2 Design Families
- Overview
- Central Processing Unit
- Graphical Processing Unit
- Field-Programmable Gate Array [7]
- System on Chip

3 Design Flow

ASYNCHRONOUS CIRCUITS

4 Linear pipeline or digital Fordism

5 Clock-driven

6 Muller Pipeline

7 Bundled-Data

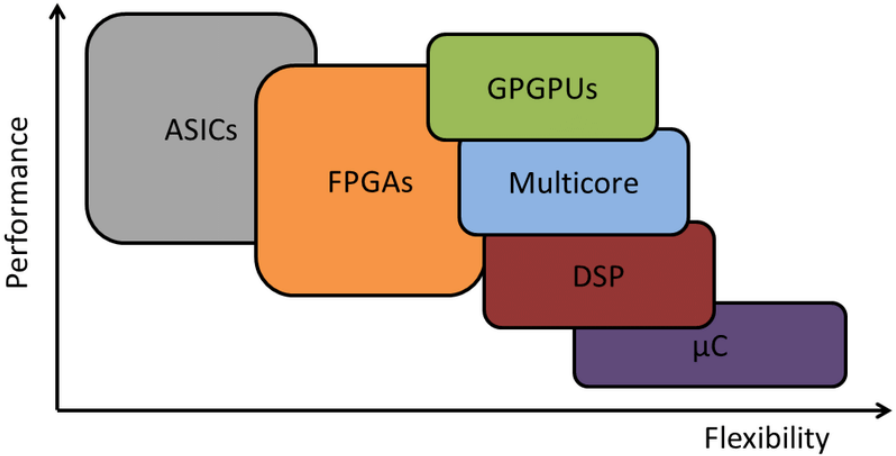


Figure: Performance vs Flexibility [1]

- Instruction Set Architecture (ISA)
- Single Instruction Single Data (SISD)
- Executes a program stored in memory

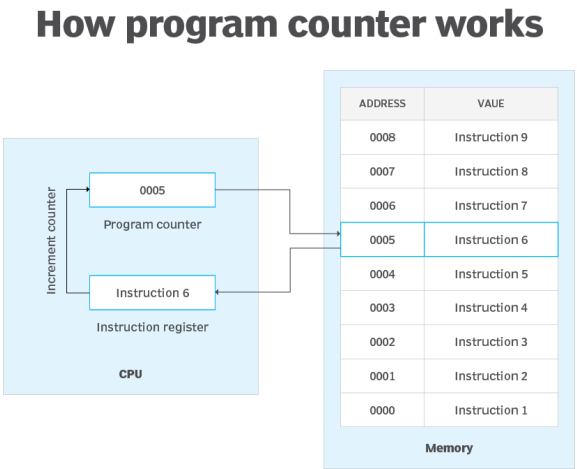


Figure: Program [2]

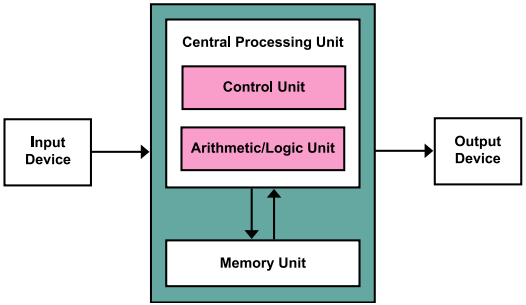


Figure: Von Neumann Architecture [3]

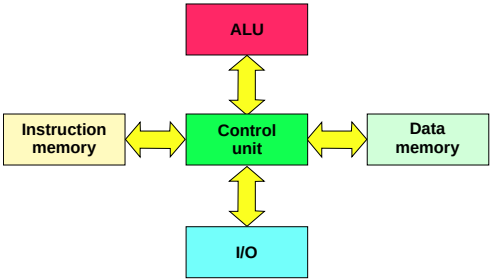


Figure: Harvard Architecture [4]

Stages :

- IF: Instruction Fetch
- ID: Instruction Decode
- EX: Execute
- MEM: Memory access
- WB: WriteBack

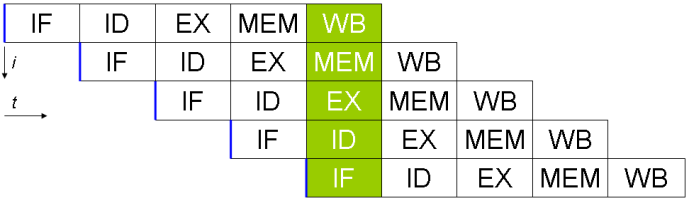


Figure: Five stage pipeline [5]

- Single Instruction Multiple Data (SIMD)
- Vector operations
  - e.g. scalar multiplication $a\vec{v}$

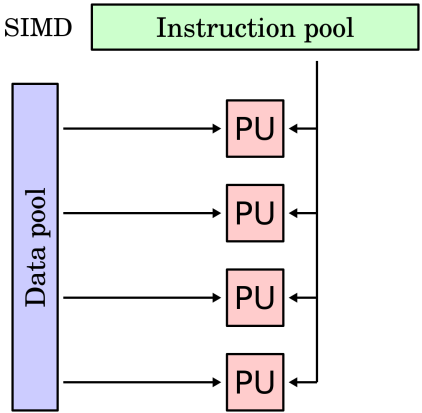
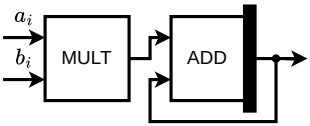


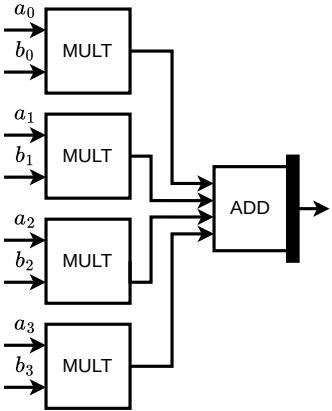
Figure: Instruction to Processing Unit (PU) [6]

Multiply ACcumulation (MAC) operation in dot product $\vec{a} \cdot \vec{b}$

Serial:

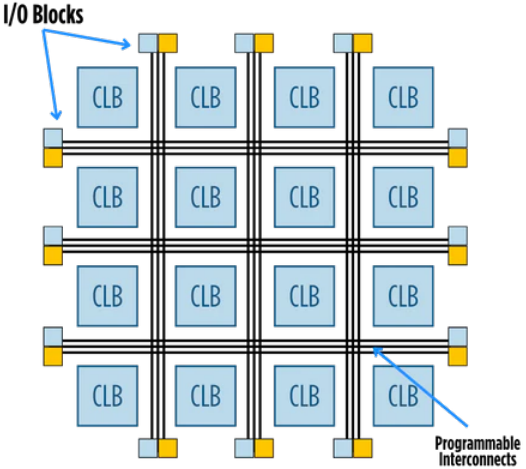


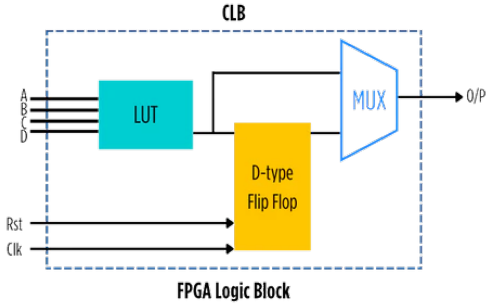
Parallel:



Comparison

- Size
- Power consumption
- Latency


I/O Blocks
CLB
Programmable
Interconnects


CLB
A
B
C
D
LUT
MUX
O/P
D-type
Flip Flop
Rst
Clk
FPGA Logic Block

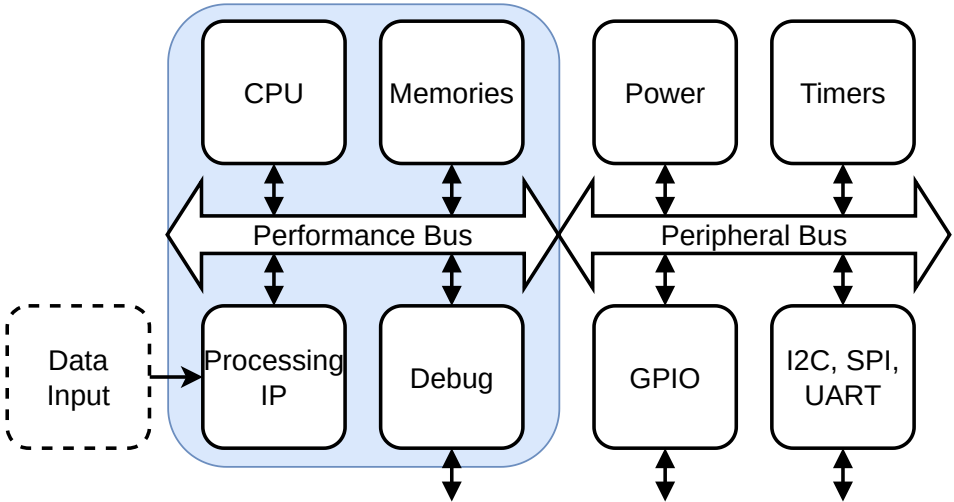

CPU
Memories
Power
Timers
Performance Bus
Peripheral Bus
Data Input
Processing IP
Debug
GPIO
I2C, SPI, UART

Table of contents

Design Flow

INTEGRATED CIRCUITS

1. Abstractions
2. Design Families
3. Design Flow
   - Successive Refinement
   - Verification & Validation

ASYNCHRONOUS CIRCUITS

4. Linear pipeline or digital Fordism
5. Clock-driven
6. Muller Pipeline
7. Bundled-Data

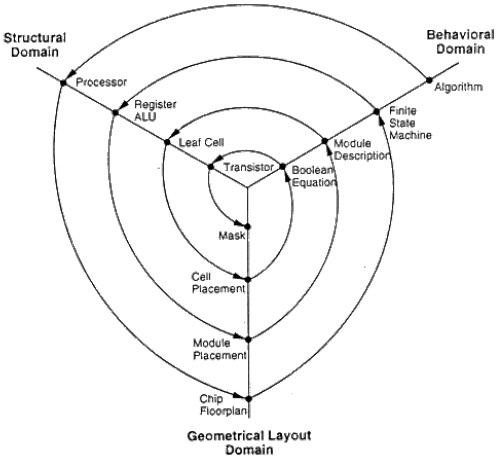


Figure: Y Chart [8]

Perspectives

- Behavioral
- Structural
- Physical/Geometry

Logical synthesis : from behavioral to structural
Physical synthesis / Layout : from structural to physical / geometrical

High Level Synthesis

- From a system-level description
- Produces a Finite State Machine (FSM)
- Followed by logical synthesis

Logical synthesis

- Takes a RTL description (e.g. FSM) & technology node
- Builds a abstract representation of the logic and maps it to standard cell
- Produces a netlist / gate-level circuit

Steps :

- Partitioning
- Floorplanning
- Placement
- Global routing
- Detailed routing

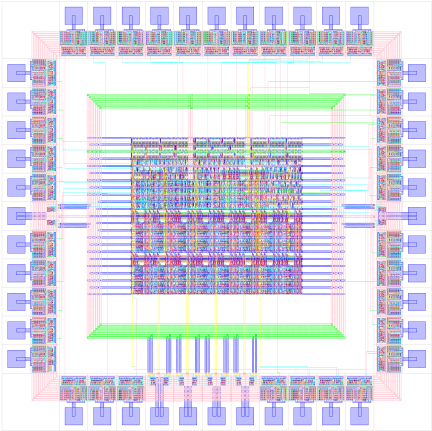


Figure: Chip layout [9]

Verification

- Functional Simulation
  - Behavioral
  - Gate-level (Timed)
  - Analog
- Formal Verification

Validation

- FPGA (Behavioral)
- Static Timing Analysis and Timing Closure

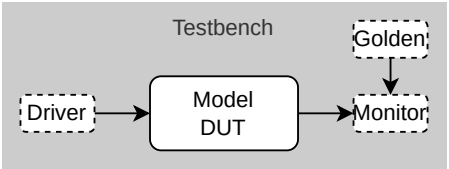


Figure: Simulated Testbench

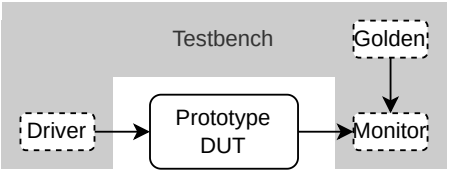


Figure: Hardware in the loop

Summary - Integrated Circuits

- The IC design flow consist of successive refinement accross abstraction layers
- Design choices : Flexibility vs Performance, Serial vs Parallel, etc.
- CPUs and GPUs are instruction set architectures that execute programs
- System on chip are composed of multiple subsystems (IPs)
- FPGA are a good platform to prototype hardware designs

Part II - ASYNCHRONOUS CIRCUITS

GUT meeting 20/01/2026

INTEGRATED CIRCUITS

1 Abstractions

2 Design Families

3 Design Flow

ASYNCHRONOUS CIRCUITS

4 Linear pipeline or digital Fordism
- Concepts

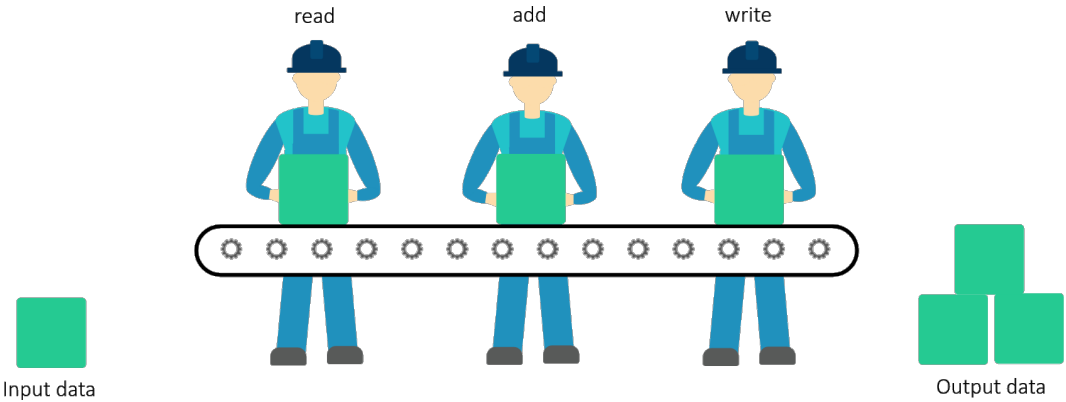
5 Clock-driven

6 Muller Pipeline

7 Bundled-Data

Assembly Line

Linear pipeline or digital Fordism

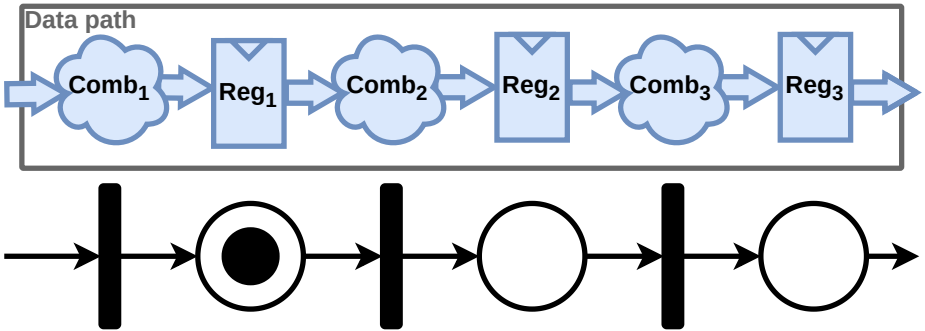


Combinational Logic

- Time-independent
- Output function of the present input

Registers

- Sequential logic
- Flip-Flop, Latch

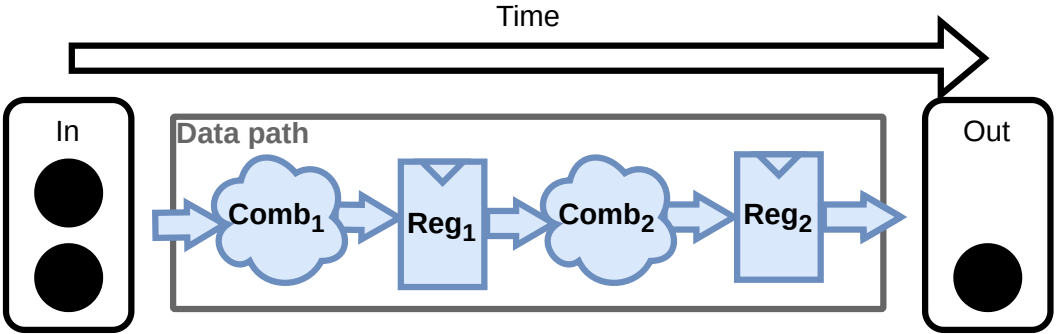


Latency

- Time from input to output

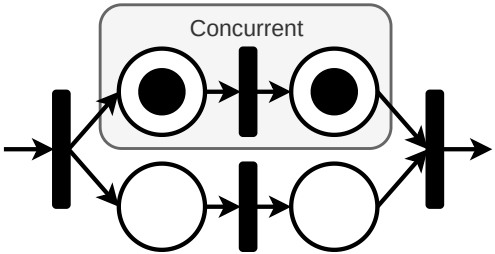
Throughput

- Number of data per time



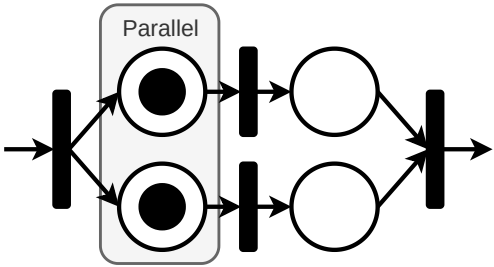
Concurrency

- Multiple stages at the same time



Parallelism

- Multiple branch at the same time



INTEGRATED CIRCUITS

1 Abstractions

2 Design Families

3 Design Flow

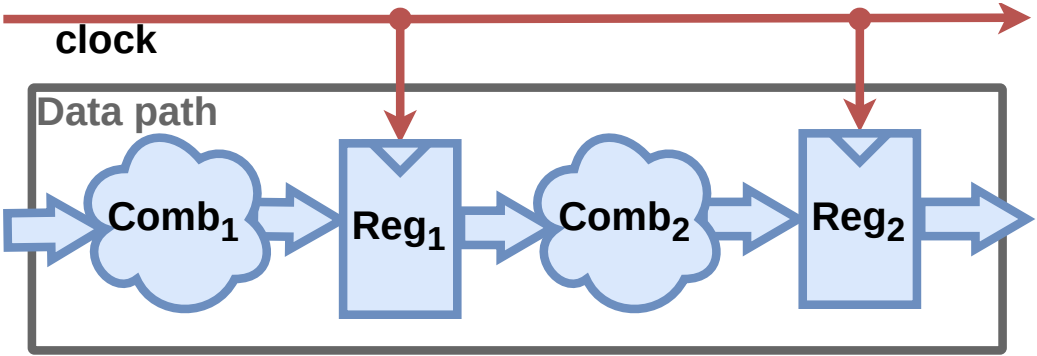
ASYNCHRONOUS CIRCUITS

4 Linear pipeline or digital Fordism

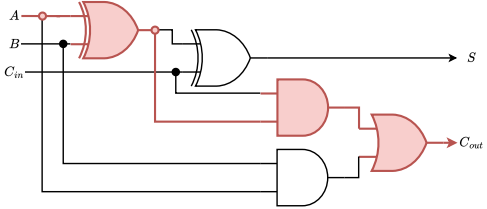
5 Clock-driven
- Timing Constraints
- Drawbacks

6 Muller Pipeline

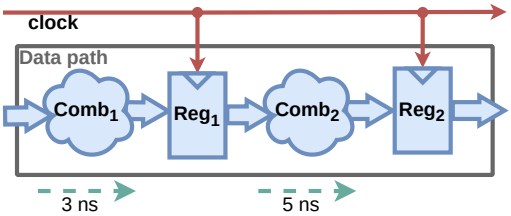
7 Bundled-Data

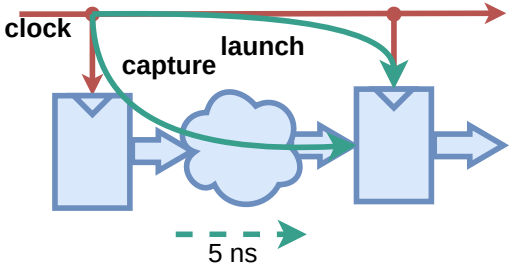

clock
Data path
Comb1
Reg1
Comb2
Reg2

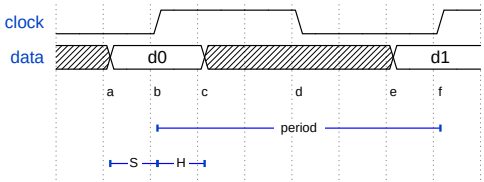
Critical Path



Timing of combinational logic

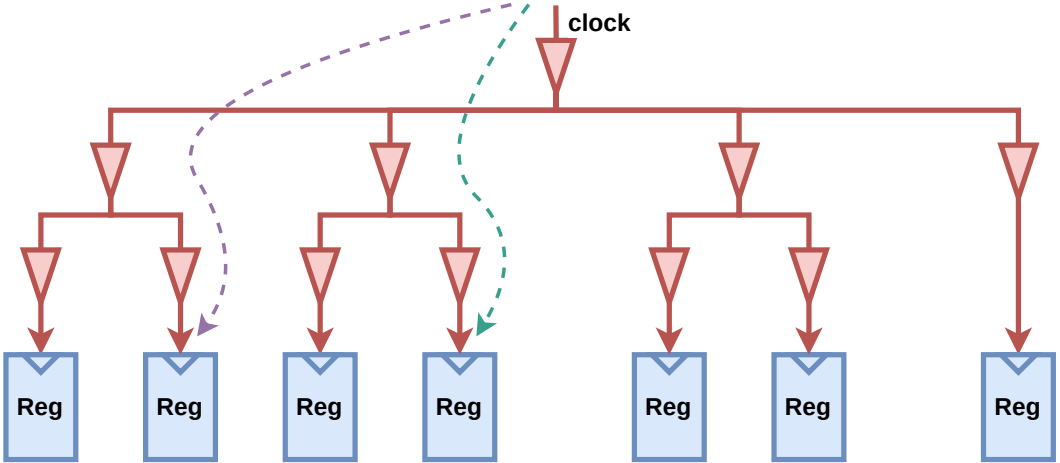




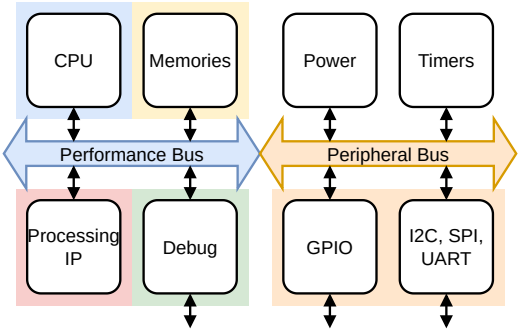


Constraints for all stages

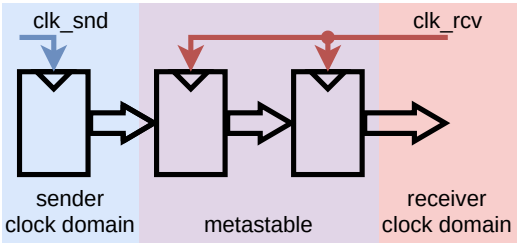
- Setup: Slack $S \geq 0$
- Hold: Memorization time H


clock
Reg
Reg
Reg
Reg
Reg
Reg
Reg

Globally Asynchronous Locally Synchronous (GALS)



Clock Domain Crossing (CDC)



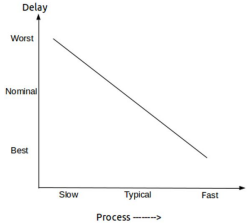
Problematic

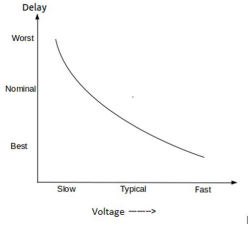
The delay of a cell is mainly influenced by three factors :

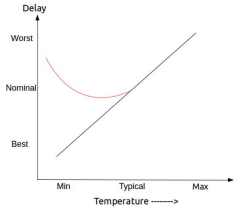
- Process variation
- Supply voltage
- Operating temperature

Solution :

- Study the circuit at different corners







INTEGRATED CIRCUITS

1 Abstractions

2 Design Families

3 Design Flow

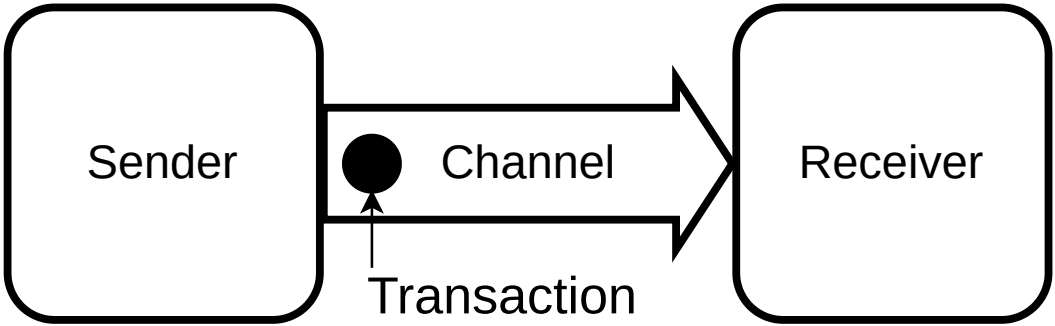
ASYNCHRONOUS CIRCUITS

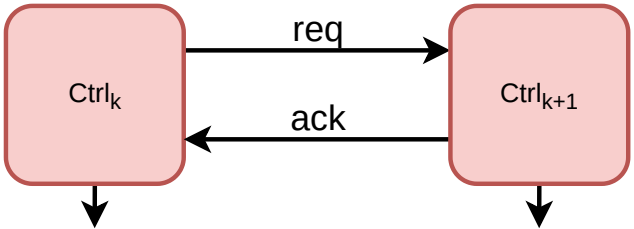
4 Linear pipeline or digital Fordism

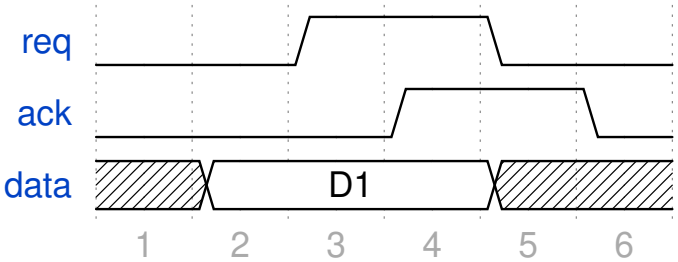
5 Clock-driven

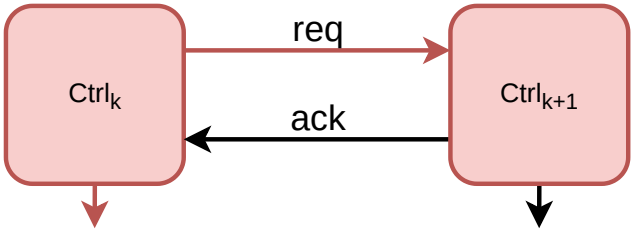
6 Muller Pipeline
- Transaction
- Signal Transition Graph
- Pipeline

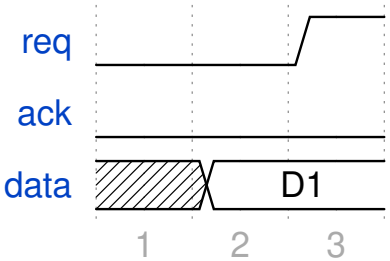
7 Bundled-Data

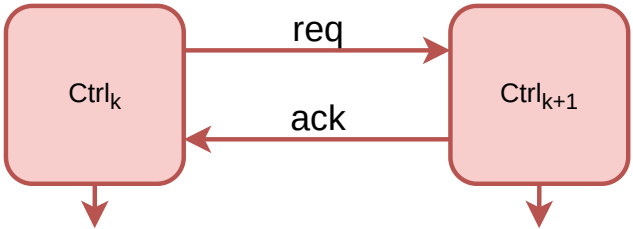

Sender
Channel
Receiver
Transaction

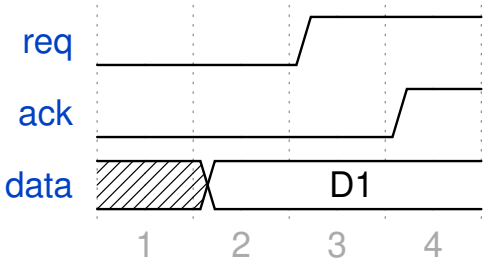

req
ack
Ctrl$_k$
Ctrl$_{k+1}$

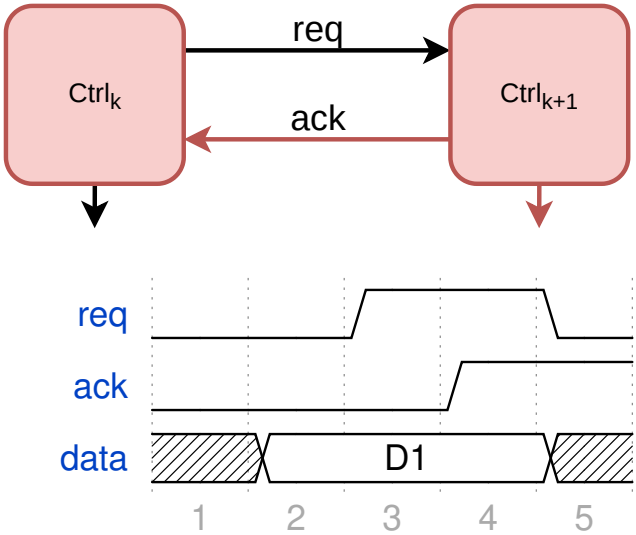

req
ack
data
D1
1
2
3
4
5
6

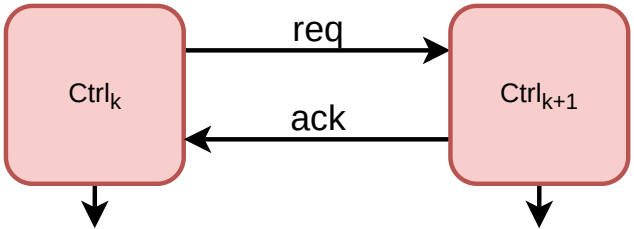

req
ack
Ctrl$_k$
Ctrl$_{k+1}$

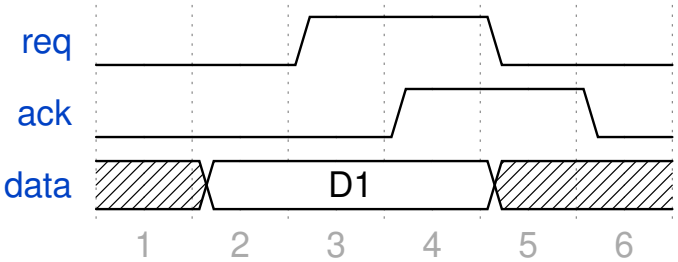

req
ack
data
D1
1
2
3

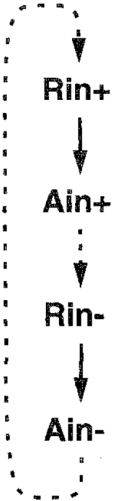

req
ack
Ctrl$_k$
Ctrl$_{k+1}$

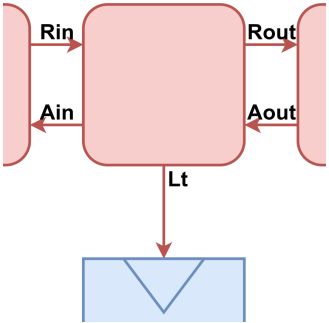

req
ack
data
D1
1
2
3
4


req
ack
Ctrl$_k$
Ctrl$_{k+1}$
req
ack
data
D1
1
2
3
4
5


req
ack
Ctrl$_k$
Ctrl$_{k+1}$

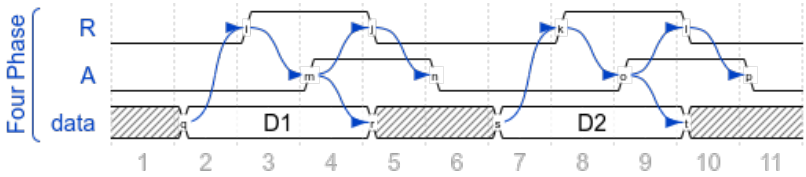

req
ack
data
D1
1
2
3
4
5
6





Signal Transition Graph

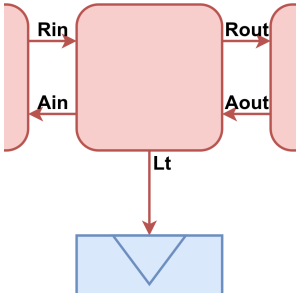
- +/-: Rising or falling edge
- arrow: internal transition
- dotted arrow: external transition

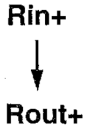




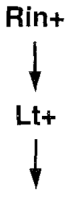
(a)

(a): request propagation



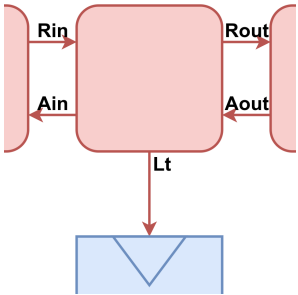


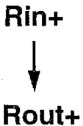
(a)



(b)

(a): request propagation
(b): data storage before acknowledge

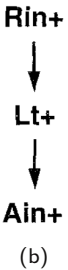




(a)

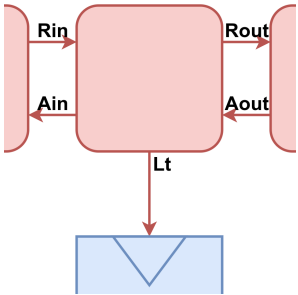


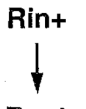
(c)



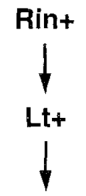
(b)

(a): request propagation
(b): data storage before acknowledge
(c): data acknowledge

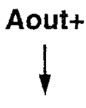




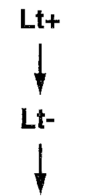
(a)



(b)

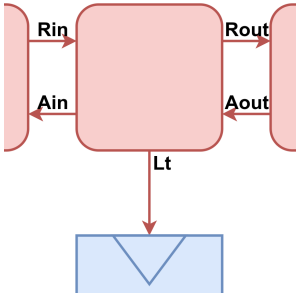


(c)



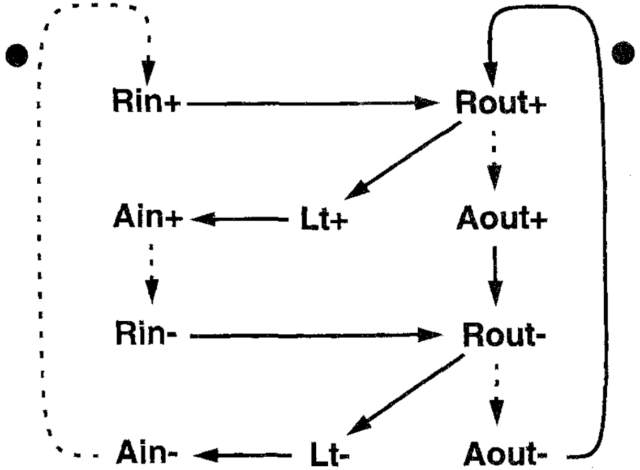
(d)

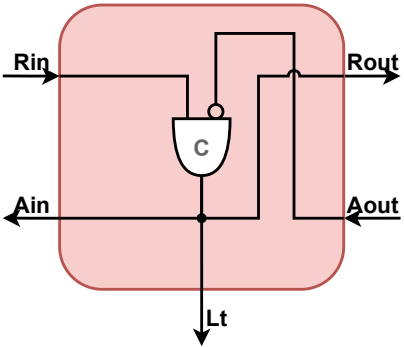
(a): request propagation
(b): data storage before acknowledge
(c): data acknowledge
(d): alternating Enable

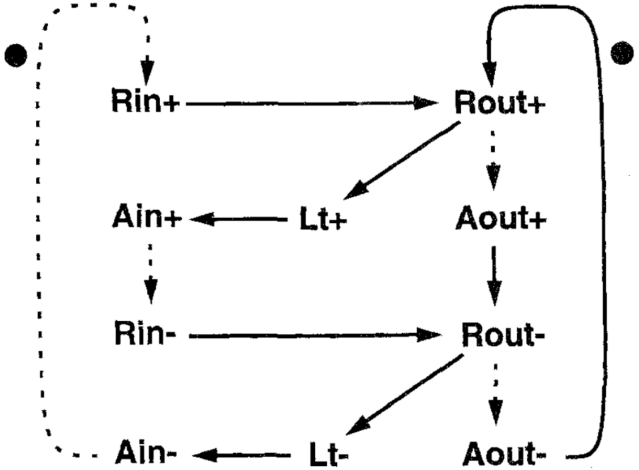


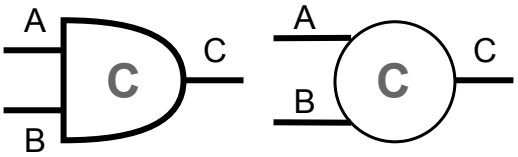
Weak Condition Half Buffer (WCHB) [11]

Muller Pipeline - Signal Transition Graph

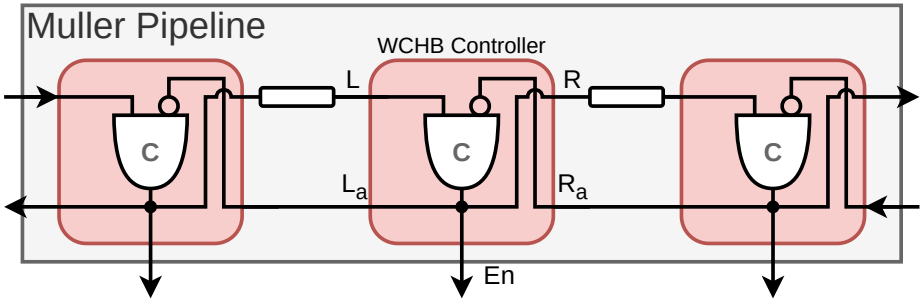



Rin
Rout
C
Ain
Aout
Lt


Rin+
Rout+
Ain+
Lt+
Aout+
Rin-
Rout-
Ain-
Lt-
Aout-



A	B	C_k
0	0	0
0	1	C_{k-1}
1	0	C_{k-1}
1	1	1



INTEGRATED CIRCUITS

1 Abstractions

2 Design Families

3 Design Flow

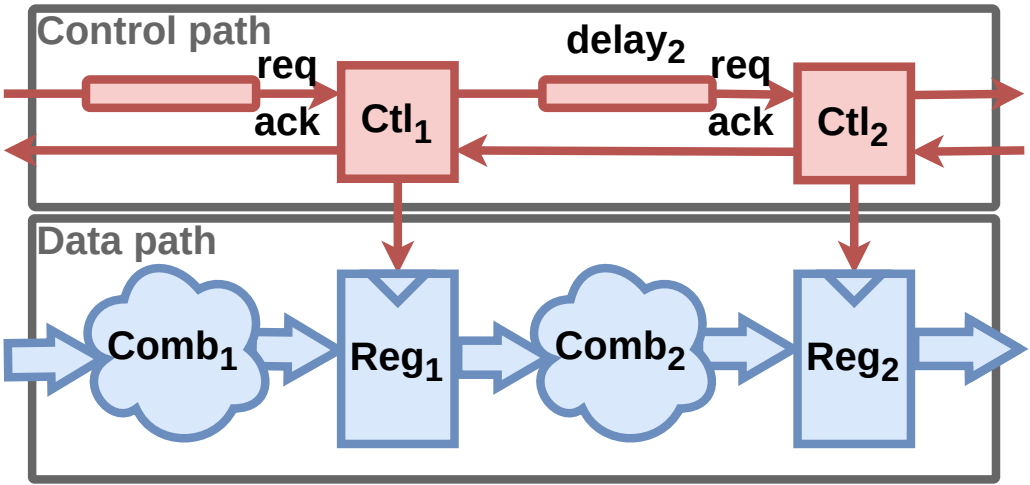
ASYNCHRONOUS CIRCUITS

4 Linear pipeline or digital Fordism

5 Clock-driven

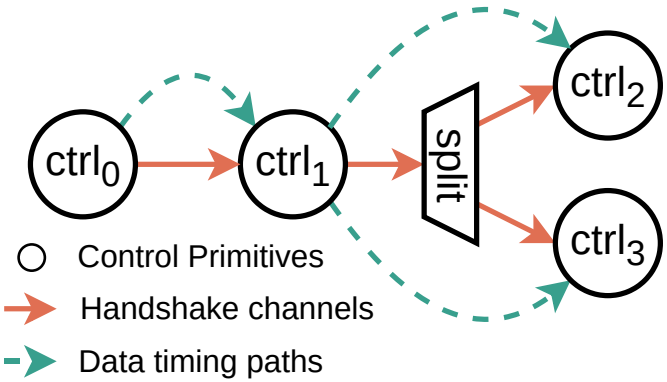
6 Muller Pipeline

7 Bundled-Data
- Control Flow
- Timing Constraints


Control path
req
ack
Ctl1
delay2
req
ack
Ctl2
Data path
Comb1
Reg1
Comb2
Reg2

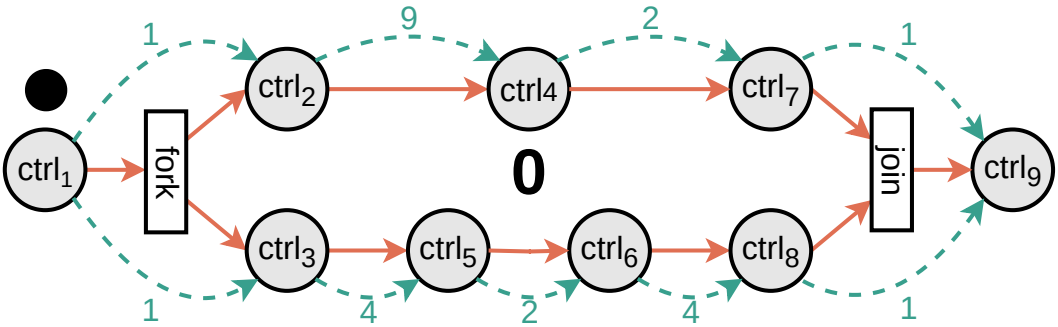
Control Primitives

- Controllers
- Fork & Join
- Split, Merge & Arbiter
- Source & Sink



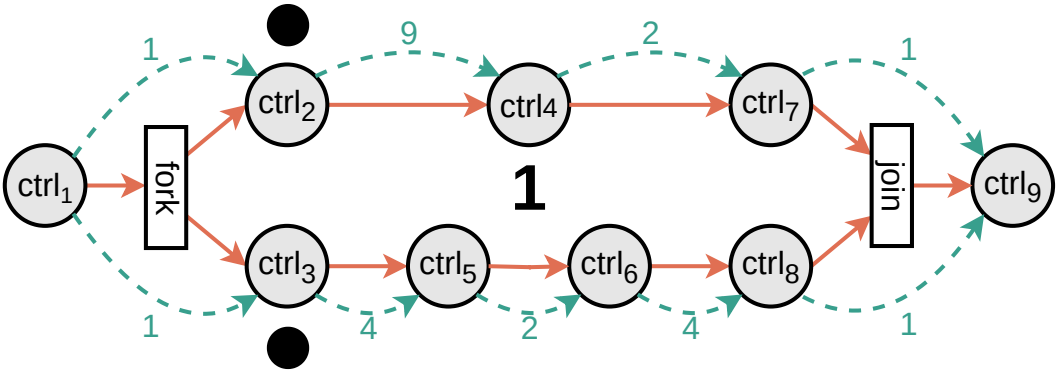
Fork & Join

Bundled-Data - Control Flow



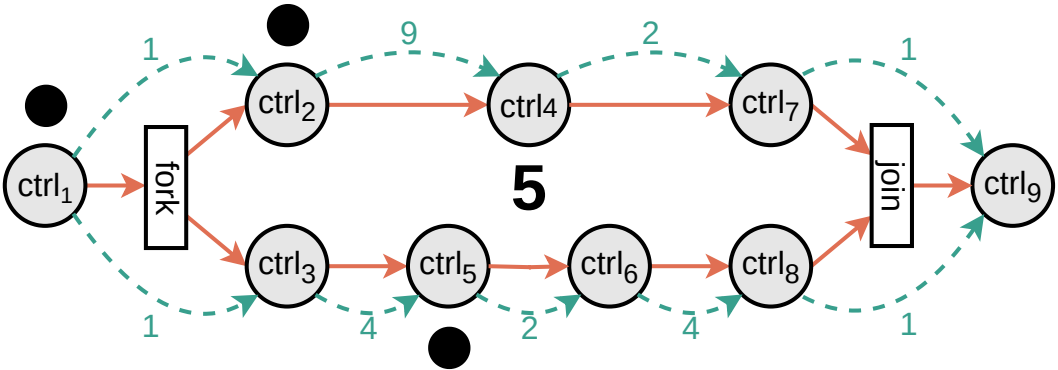
Fork & Join

Bundled-Data - Control Flow



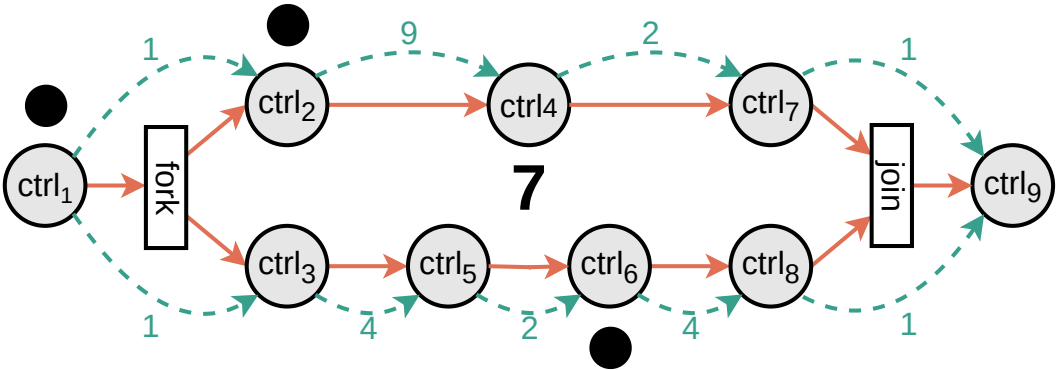
Fork & Join

Bundled-Data - Control Flow



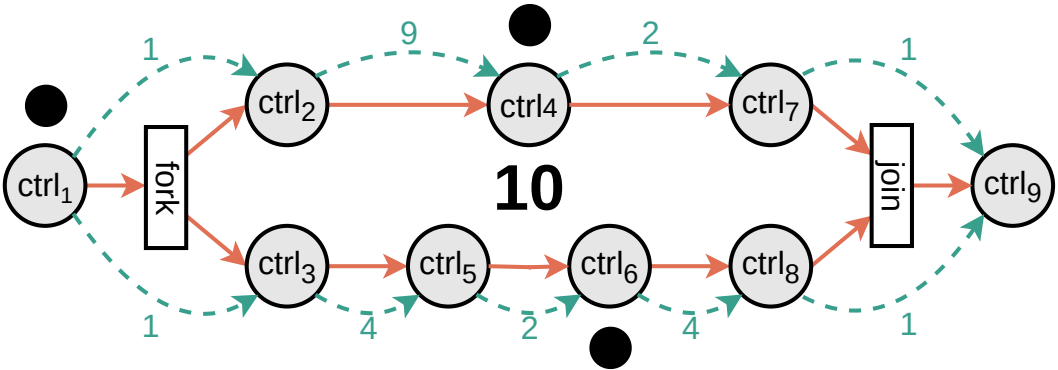
Fork & Join

Bundled-Data - Control Flow



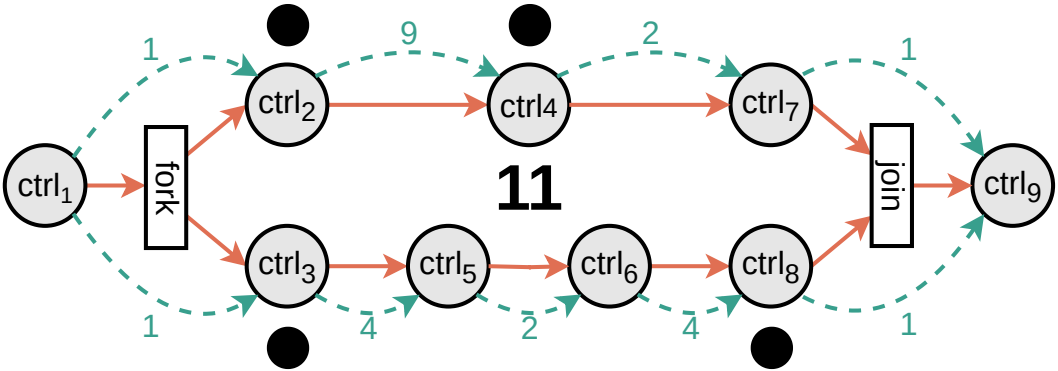
Fork & Join

Bundled-Data - Control Flow



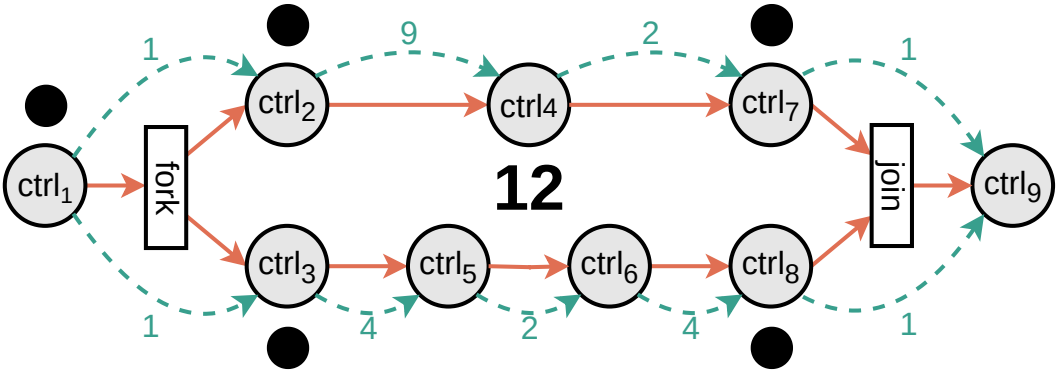
Fork & Join

Bundled-Data - Control Flow



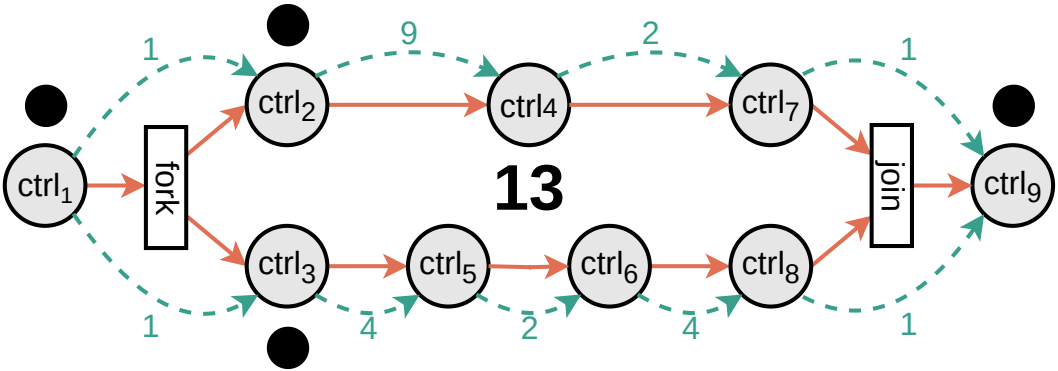
Fork & Join

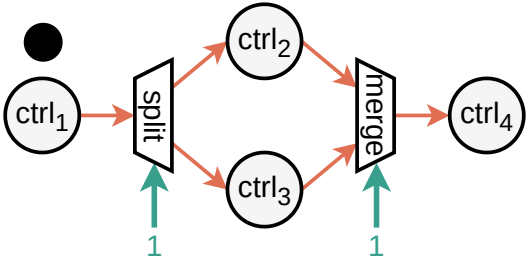
Bundled-Data - Control Flow

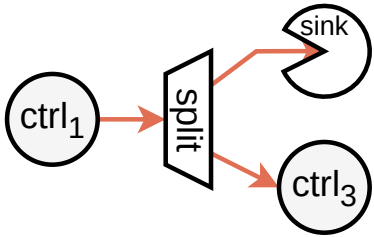


Fork & Join

Bundled-Data - Control Flow

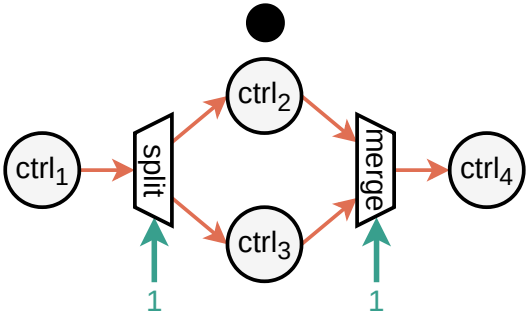


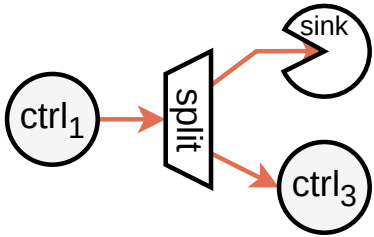




Advantages

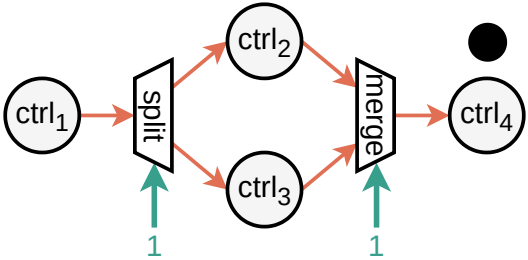
- Data-driven
- Partial activation of the logic
- Event-gating

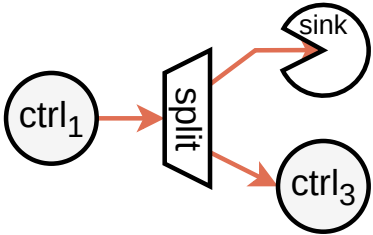




Advantages

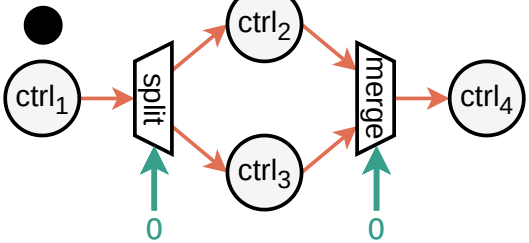
- Data-driven
- Partial activation of the logic
- Event-gating

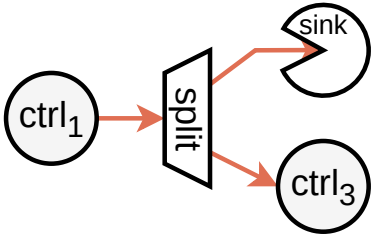




Advantages

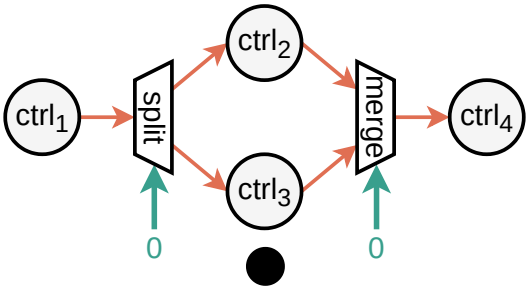
- Data-driven
- Partial activation of the logic
- Event-gating

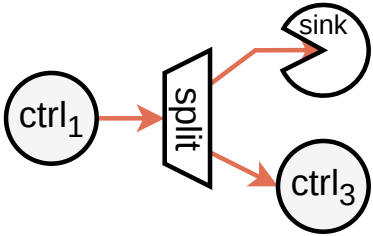




Advantages

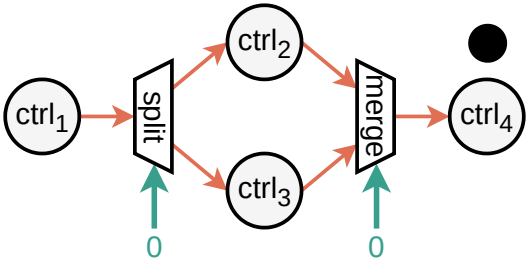
- Data-driven
- Partial activation of the logic
- Event-gating

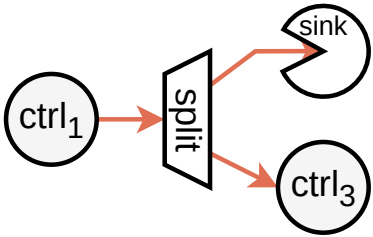




Advantages

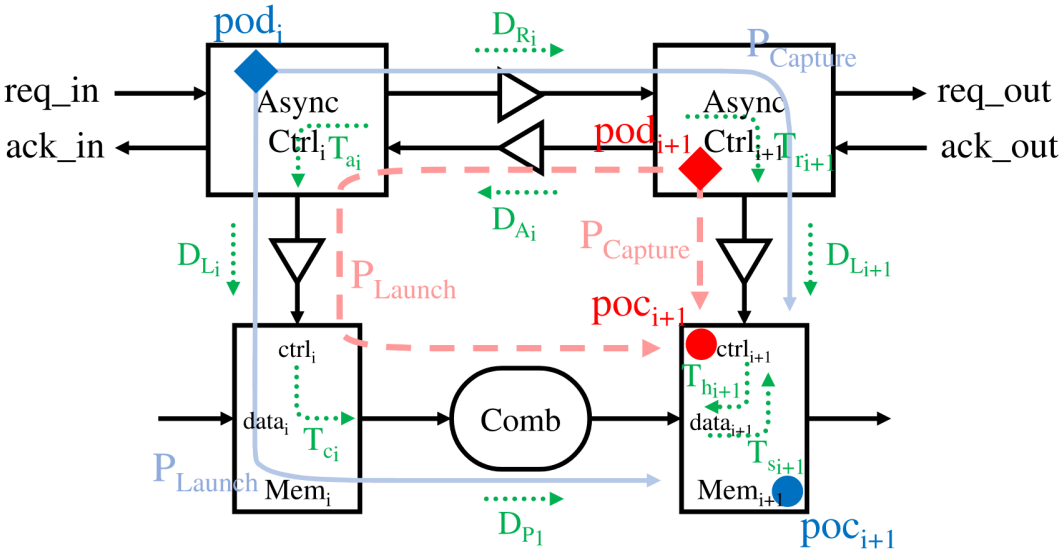
- Data-driven
- Partial activation of the logic
- Event-gating

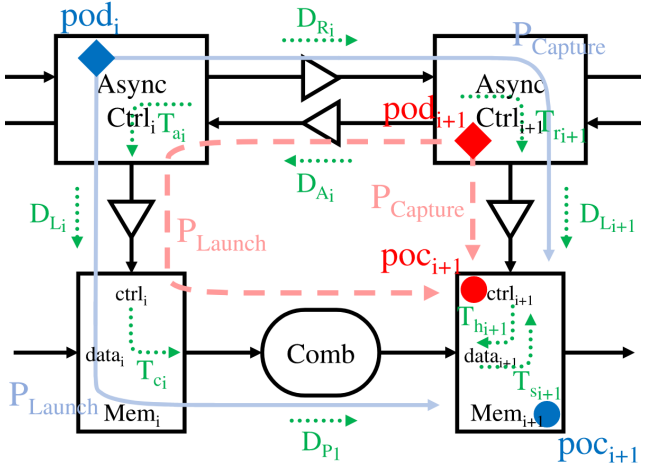




Advantages

- Data-driven
- Partial activation of the logic
- Event-gating


pod_i
D_Ri
P_Capture
req_in
ack_in
Async
Ctrl_i
T_ai
Async
Ctrl_i+1
T_ri+1
pod_i+1
req_out
ack_out
D_Ai
D_Li
P_Launch
P_Capture
D_Li+1
poc_i+1
ctrl_i
ctrl_i+1
T_hi+1
data_i
T_ci
Comb
data_i+1
T_si+1
P_Launch
Mem_i
Mem_i+1
D_Pl
poc_i+1



Relative Timing Constraint

$$pod_i \mapsto data_{i+1} \prec ctrl_{i+1}$$

Delay Equation

$$D_{R_i} + T_{r_{i+1}} + D_{L_{i+1}} > D_{L_i} + T_{c_i} + D_{P_i} + T_{s_{i+1}}$$

Matched Delay : D_{R_i}

Summary - Asynchronous Circuits

- Data travels from register to register
- The data path consists of sequential and combinational logic
- Data is propagated through a global clock in synchronous circuits
- In asynchronous circuits the propagation is local and transactionnal
- In Bundled-Data, transactions are done via handshakes between controllers
- All digital circuits are subject to variability issues, requiring extensive timing analysis
- Asynchronous Circuits are advantageous for the following reasons :
  - Lower power consumption
  - More robustness and modularity
  - Less timing constraints

Table of contents

8 References

9 Oligopoly problem in VLSI

10 Metastability & Arbiters

11 Properties of Petri Nets & STG

12 Families of Bundled-Data circuits

References I

[1] R. Molanes, K. Amarasinghe, J. Rodriguez-Andina, and M. Manic, "Deep learning and reconfigurable platforms in the internet of things: Challenges and opportunities in algorithms and hardware," *IEEE Industrial Electronics Magazine*, vol. 12, pp. 36–49, Jun. 2018. DOI: 10.1109/MIE.2018.2824843

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Electronic Design Automation tools

- Synopsys (Fusion Design Platform)
- Cadence (Virtuoso and Spectre)
- Siemes (aka Mentor Graphics)

Foundries [13] & Process Design Kit

- Asia :
  - TSMC (1)
  - Samsung (2)
  - UMC (5)
  - SMIC (6)
- USA :
  - Intel (3)
  - Global Foundries (4)
  - Texas Instruments (7)
- Europe :
  - Infineon (8)
  - NXP (9)
  - STMicroelectronics (10)

"The OpenROAD flow delivers an Autonomous, No-Human-In-Loop (NHIL) flow, 24 hour turnaround from RTL-GDSII for rapid design exploration and physical design implementation."

Tools

- Yosys ABC
- OpenSTA
- TritonRoute

Process Design Kit

- Google Skywater 130nm
- Global Foundries 180nm

Tools

- Alliance/Coriolis [9]

Process Design Kit

- IHP Open130-G2

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Occupancy & Static Spread

Families of Bundled-Data circuits

Capture

Families of Bundled-Data circuits

Latched Controller

Families of Bundled-Data circuits